

Question

In nature, potassium has three isotopes, ^{39}K , ^{40}K and ^{41}K , with abundances of 93.2581%, 0.0117% and 6.7302% respectively. ^{39}K and ^{41}K are stable nuclides; while ^{40}K is radioactive, and its decay may produce ^{40}Ca (partial half-life $t_1 = 1.397 \times 10^9 \text{ y}$) or ^{40}Ar (partial half-life $t_2 = 1.193 \times 10^{10} \text{ y}$). Argon has three isotopes, ^{36}Ar , ^{38}Ar and ^{40}Ar , with abundances of 0.334%, 0.063% and 99.604% respectively. ^{36}Ar , ^{38}Ar and ^{40}Ar are all stable nuclides. Granite is a rock formed by the cooling and solidification of magma from volcanic eruptions. When granite forms, it seals a very small amount of atmosphere inside the rock. These gases cannot diffuse out of the rock unless they are completely released by heating and melting at high temperatures. To determine the age of a piece of granite, a 1.000 g granite sample was thoroughly heated and melted to completely release the argon gas inside; the released gas was mixed with $n_1 = 8.30 \times 10^{-10} \text{ mol}$ of pure ^{38}Ar , and the mass spectrometer measured $n(^{38}\text{Ar}) : n(^{36}\text{Ar}) = 371.8$ and $n(^{40}\text{Ar}) : n(^{38}\text{Ar}) = 1.400$ in the mixed gas. Another sample was taken, and atomic absorption spectroscopy was used to measure the potassium content as 6.065%. Assume that the isotopic abundances of potassium and argon in nature do not change with age, and the following options are closest to the age of the granite sample

A. $1 \times 10^3 \text{ y}$

B. $5 \times 10^3 \text{ y}$

C. $1 \times 10^4 \text{ y}$

D. $5 \times 10^4 \text{ y}$

E. $1 \times 10^5 \text{ y}$

F. $5 \times 10^5 \text{ y}$

G. $1 \times 10^6 \text{ y}$

H. 5×10^6 y

I. 1×10^7 y

J. 5×10^7 y

K. 1×10^8 y

L. 5×10^8 y

M. 1×10^9 y

N. 5×10^9 y

O. 1×10^{10} y

Answer

Correct Answer: **J**

Detailed Explanation

$n(^{36}\text{Ar})$ is a stable nuclide that can react to sequester the total amount of Ar in the gas. Suppose the total amount of Ar in the sequestered gas is x mol, then we have the equation:

$$\frac{0.00063x + 8.30 \times 10^{-10}}{0.00334x} = 371.8$$

Solving for $x = 6.69 \times 10^{-10}$, the total amount of Ar is 6.69×10^{-10} mol

CHECKPOINT

1 PTS

The total amount of Ar is 6.69×10^{-10} mol

Since $n(^{38}\text{Ar}) = 8.30 \times 10^{-10}$ mol, we can calculate $n(^{40}\text{Ar}) = 1.4n(^{38}\text{Ar}) = 1.16 \times 10^{-9}$ mol

From the isotopic abundance, $n(^{40}\text{Ar})_1$ from Ar itself is $0.99604x = 6.67 \times 10^{-10}$ mol

CHECKPOINT

1 PTS

$n(^{40}\text{Ar})_1$ from Ar itself is $0.99604x = 6.67 \times 10^{-10}$ mol

Therefore, $n(^{40}\text{Ar})_2$ from ^{40}K is $n(^{40}\text{Ar}) - n(^{40}\text{Ar})_1 = 4.93 \times 10^{-10}$ mol

CHECKPOINT

1 PTS

$$n(^{40}\text{Ar})_2 \text{ from } ^{40}\text{K} \text{ is } n(^{40}\text{Ar}) - n(^{40}\text{Ar})_1 = 4.93 \times 10^{-10} \text{ mol}$$

According to atomic absorption spectroscopy, and using 39.10 to approximate the weighted average atomic mass of potassium, we can calculate the total amount of K, $n(\text{K}) = \frac{1.000 \times 0.06065}{39.10} = 0.001551 \text{ mol}$

CHECKPOINT

1 PTS

$$\text{Total K, } n(\text{K}) = \frac{1.000 \times 0.06065}{39.10} = 0.001551 \text{ mol}$$

$$\text{Then } n(^{40}\text{K}) = 0.000117n(\text{K}) = 1.81 \times 10^{-7} \text{ mol}$$

CHECKPOINT

1 PTS

$$n(^{40}\text{K}) = 0.000117n(\text{K}) = 1.81 \times 10^{-7} \text{ mol}$$

The decay reaction is a first-order reaction, and the rate constant k can be obtained from the half-life:

$$k_1 = \frac{\ln 2}{t_1} = 4.962 \times 10^{-10} \text{ y}^{-1}$$

CHECKPOINT

1 PTS

$$k_1 = \frac{\ln 2}{t_1} = 4.962 \times 10^{-10} \text{ y}^{-1}$$

$$k_2 = \frac{\ln 2}{t_2} = 5.810 \times 10^{-11} \text{ y}^{-1}$$

CHECKPOINT

1 PTS

$$k_2 = \frac{\ln 2}{t_2} = 5.810 \times 10^{-11} \text{ y}^{-1}$$

Then the consumption rate of ^{40}K can be expressed as:

$$\frac{-dn(^{40}\text{K})}{dt} = k_1 n(^{40}\text{K}) + k_2 n(^{40}\text{K}) = (k_1 + k_2) n(^{40}\text{K})$$

Integrating, we have $\ln \frac{n(^{40}\text{K})}{n(^{40}\text{K})_{\text{end}}} = (k_1 + k_2)t$

CHECKPOINT

1 PTS

$$\ln \frac{n(^{40}\text{K})}{n(^{40}\text{K})_{\text{end}}} = (k_1 + k_2)t$$

Also, the total amount of ^{40}K consumed is $n(^{40}\text{Ar})_2 \times \frac{k_1 + k_2}{k_2} = 4.70 \times 10^{-9} \text{ mol}$

CHECKPOINT

1 PTS

The total amount of ^{40}K consumed is $n(^{40}\text{Ar})_2 \times \frac{k_1 + k_2}{k_2} = 4.70 \times 10^{-9} \text{ mol}$

Then $n(^{40}\text{K})_{\text{end}} = n(^{40}\text{K}) - 4.70 \times 10^{-9} = 1.76 \times 10^{-7} \text{ mol}$

Substituting each value into $\ln \frac{n(^{40}\text{K})}{n(^{40}\text{K})_{\text{end}}} = (k_1 + k_2)t$, solving for $t = 4.74 \times 10^7 \text{ y}$

CHECKPOINT

1 PTS

$$t = 4.74 \times 10^7 \text{ y}$$